

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4

Total Marks:40

Annexure A

ARTICLE REVIEW

Code of Conduct:

I **B.SREENITHA (192312203)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

B. SREENITHA

Signature of the student:

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications*.

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

- (a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.
- (b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

- (a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.
- (b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

- (a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.
- (b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?
- (c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

- (a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.
 - (b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.
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RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

CO4 ASSESSMENT TOOL 2 – ARTICLE REVIEW

Selected Research Article

Article: *Prediction of diabetes using logistic regression and ensemble techniques*

Authors: [Article authors as listed by Elsevier]

Journal: *Computer Methods and Programs in Biomedicine Update*

Year: 2021

Volume: 1

Article number: 100032

DOI: 10.1016/j.cmpbup.2021.100032

Publisher: Elsevier

Domain: Healthcare / Medical Diagnosis

ML Topic: Statistical Learning – Logistic Regression and Ensemble Learning

The article uses Logistic Regression as the main classification technique and investigates whether preprocessing, feature selection and ensemble methods can improve diabetes prediction.

TASK 1: ARTICLE SUMMARY

(a) Citation, Domain and ML Problem

Full Citation

Prediction of diabetes using logistic regression and ensemble techniques. *Computer Methods and Programs in Biomedicine Update*, Volume 1, 2021, Article 100032.

DOI: **10.1016/j.cmpbup.2021.100032.**

Domain/Application Area

The article belongs to the **healthcare and medical diagnosis** domain. It focuses on using machine learning to predict whether a patient has diabetes based on medical and diagnostic measurements.

Specific ML Problem

The ML problem is a **binary classification problem**:

- **Class 0:** Patient does not have diabetes.
- **Class 1:** Patient has diabetes.

The study investigates Logistic Regression and ensemble methods to improve prediction performance.

(b) Article Objective – 150–200 Word

The main objective of the article is to develop an effective machine-learning-based model for predicting diabetes from clinical data. The authors focus primarily on Logistic Regression because it is a classification method that provides probabilistic estimates and is commonly used for clinical analysis. The study investigates whether appropriate preprocessing, feature selection and ensemble learning can improve the performance of a basic Logistic Regression model. Two datasets are considered: the PIMA Indians Diabetes dataset from the National Institute of Diabetes and Digestive and Kidney Diseases and a second dataset from Vanderbilt involving rural African American patients in Virginia. The authors first preprocess the data by handling null and missing values and then apply feature-selection techniques. For one dataset, new features are generated and correlation analysis is used to identify useful variables, while chi-square-based feature selection is applied to the second dataset. Ensemble approaches, including Max Voting and Stacking, are then

investigated. The research addresses the limitation that the choice of algorithm alone is not sufficient for accurate prediction. It shows that preprocessing, feature selection, cross-validation and ensemble techniques can improve diabetes prediction performance.

TASK 2: METHODOLOGY ANALYSIS

(a) ML Techniques and Dataset

Machine Learning Techniques

The primary technique used in the article is **Logistic Regression**.

Logistic Regression predicts the probability of a binary outcome.

The model can be represented as:

where:

The study then investigates ensemble techniques:

1. **Max/Majority Voting**
2. **Stacking**

These methods combine predictions from multiple models to improve overall prediction performance.

Datasets

The article uses **two datasets**:

Dataset	Source	Purpose
Dataset 1	PIMA Indians Diabetes dataset	Diabetes prediction
Dataset 2	Vanderbilt study	Diabetes prediction

The PIMA dataset originates from the **National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK)**. The second dataset is based on a study of rural African Americans in Virginia.

Data Preparation

The article uses:

- Null-value preprocessing
- Missing-value removal
- Feature engineering
- Correlation analysis
- Chi-square feature selection
- Normalization
- Cross-validation

For Dataset 1, the authors create new features from existing variables and use a correlation heatmap to select eight important features.

For Dataset 2, **univariate feature selection using the chi-square test** is used to select eight top-scoring features.

(b) Mapping to CO4

The article directly maps to the **Statistical Learning Methods** component of CO4.

CO4 Topic	Application in Article
Inductive Learning	Models learn patterns from labelled patient records
Decision Trees	Not the primary method
Statistical Learning	Logistic Regression
Reinforcement Learning	Not used
Classification	Diabetes / non-diabetes prediction

The course outcome specifically includes statistical learning methods such as Logistic Regression, which makes this article relevant to CO4.

CO4_AT2.docx

Methodological Strength

A major strength is that the authors do not rely only on the basic Logistic Regression algorithm. They investigate **preprocessing, feature selection, cross-validation and ensemble methods**, providing a broader evaluation of factors affecting model performance.

Methodological Limitation

A limitation is that the study depends on existing datasets, including the PIMA dataset. Therefore, the results may not automatically generalize to all populations and healthcare environments.

TASK 3: RESULTS & CRITICAL EVALUATION

(a) Key Results

The article reports different results for the two datasets.

Dataset	Best Approach	Accuracy
PIMA Indians Diabetes	Max Voting ensemble	≈78%
Vanderbilt dataset	Max Voting / Stacking	≈93%

The authors report that the highest accuracy was approximately **78% for Dataset 1** and approximately **93% for Dataset 2** after applying ensemble techniques.

The results indicate that ensemble approaches can improve upon a single Logistic Regression model.

Main Findings

- Logistic Regression provides an effective baseline.
- Feature selection improves the model.
- Removing null and redundant values improves performance.
- Normalization is useful during preprocessing.
- Ensemble techniques improve prediction performance.
- Cross-validation helps evaluate model reliability.

(b) Critical Evaluation

Are the Results Realistic?

Yes. The reported accuracies are plausible for diabetes prediction datasets.

The difference between the two datasets also shows that model performance depends strongly on the characteristics and quality of the dataset.

However, **accuracy alone is not sufficient for a medical diagnosis system.**

Additional metrics should include:

- Precision
- Recall
- F1-score
- Specificity
- ROC-AUC
- Confusion matrix
- Calibration

Potential Biases

There are several possible sources of bias:

1. Dataset Bias

The datasets represent particular populations and may not represent every demographic group.

2. Sampling Bias

The characteristics of the patients included in the datasets may differ from the general population.

3. Evaluation Bias

Performance can vary depending on the train-test split and validation strategy.

4. Class Distribution

If diabetic and non-diabetic samples are not balanced, accuracy may give a misleading impression of model performance.

Additional Experiments

The research could be strengthened by:

1. Testing the model on an independent external dataset.
2. Applying stratified k-fold cross-validation.
3. Reporting ROC-AUC and F1-score.
4. Testing SMOTE for class imbalance.
5. Performing statistical significance testing.
6. Using explainability techniques such as SHAP.
7. Testing the model prospectively using real clinical data.

(c) Real-World Applicability

The proposed approach can potentially be deployed in:

- Hospitals
- Primary healthcare centers
- Diabetes screening systems
- Electronic Health Record systems
- Clinical decision-support systems
- Telemedicine applications
- Mobile health applications

Industry Challenges

Data Availability:

Healthcare organizations require large, high-quality and properly labelled datasets.

Data Privacy:

Patient medical information must be protected.

Model Generalization:

A model trained on one population may not perform equally well on another population.

Integration:

The model needs to integrate with existing hospital/EHR systems.

Interpretability:

Doctors need to understand why a prediction was generated.

Regulatory and Ethical Issues:

The model should support clinical decisions rather than blindly replacing medical professionals.

TASK 4: REFLECTION & LEARNING OUTCOME

(a) Three Key Insights

Insight 1 – Logistic Regression for Classification

I learned that Logistic Regression can be effectively used for **binary classification** problems such as predicting whether a patient has diabetes.

CO4 Connection: Statistical Learning Methods.

Insight 2 – Importance of Data Preprocessing

The article demonstrates that model performance does not depend only on the machine learning algorithm. **Missing-value handling, normalization and feature selection** can significantly influence the final result.

CO4 Connection: Inductive Learning and classification model preparation.

Insight 3 – Ensemble Learning

I learned that combining multiple model predictions through **Max Voting or Stacking** can improve prediction performance compared with relying on a single model.

CO4 Connection: Statistical Learning and classification.

(b) Proposed Novel Extension

Proposed Experiment

I would extend the research by developing a **bias-aware diabetes prediction model using a larger and more diverse multi-hospital dataset**.

The experiment would:

1. Combine diabetes records from multiple hospitals.
2. Apply missing-value handling.
3. Balance classes using SMOTE where appropriate.
4. Train Logistic Regression, Random Forest, SVM and ensemble models.
5. Use stratified k-fold cross-validation.
6. Evaluate Accuracy, Precision, Recall, F1-score and ROC-AUC.
7. Apply SHAP for model explainability.
8. Compare performance across different demographic groups.

Justification

This extension is feasible because the existing research already demonstrates that preprocessing, feature selection and ensemble learning can improve diabetes prediction.

Using a larger and more diverse dataset would improve **generalizability**, while SHAP-based explanations could make the model more understandable to healthcare professionals.

Expected Impact

The proposed enhancement could produce a more reliable and interpretable diabetes screening system that is better suited for real-world healthcare environments.

Conclusion

The reviewed article demonstrates that **Logistic Regression combined with proper preprocessing, feature selection and ensemble techniques** can provide effective diabetes

prediction. The study is directly relevant to CO4 because it applies statistical learning and classification concepts. The results also demonstrate an important machine-learning principle: **good performance depends not only on the algorithm but also on data quality, feature selection and evaluation methodology.**